

CHE 212

HEAT TRANSFER

**Two-Dimensional Heat
Conduction in a Metal Plate**

GROUP 16

OUR TEAM

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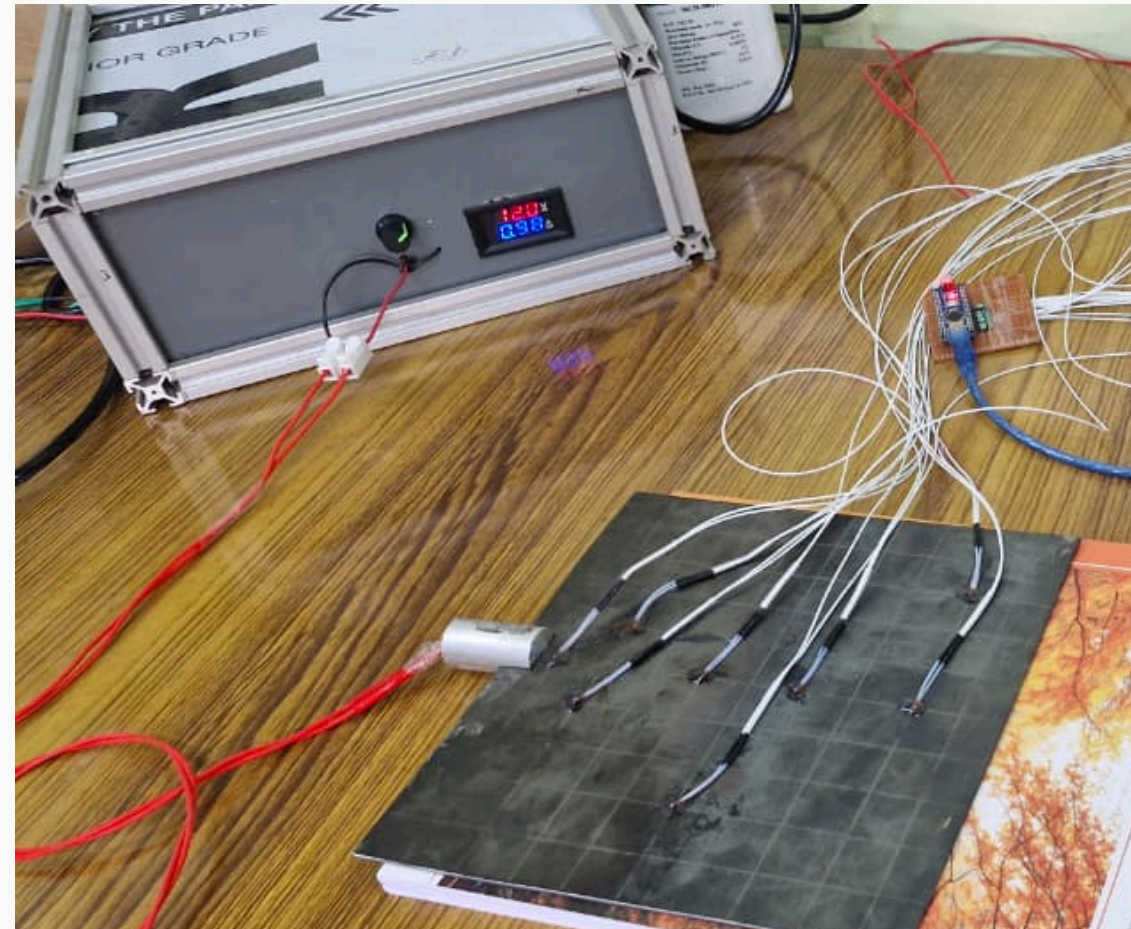
Tripti 241103

Objective

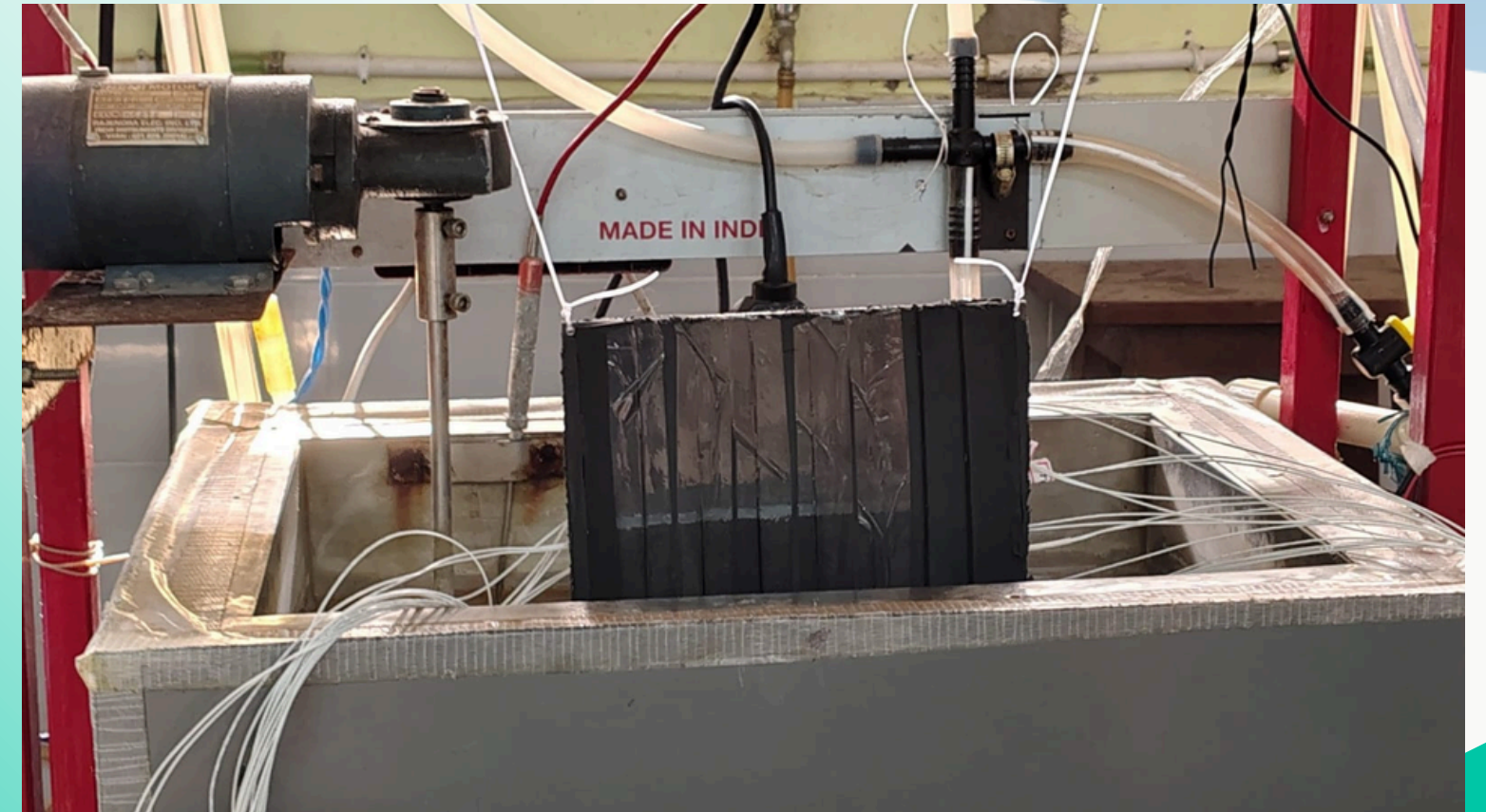
- To study steady-state heat conduction in a 2D plate.
- To measure temperature distribution experimentally using thermocouples.
- To develop a numerical model (FDM) for temperature prediction.
- To compare experimental vs simulation results.

Changes in Setup

1. Constant Heat flux as the heating source.
2. Bottom and Top surfaces are exposed to ambient air.



1. Constant temperature at the one edge (water bath).
2. Large Surfaces are insulated.



Problem Statement

- Heat transfer in a solid slab occurs through conduction within the material and heat loss to the surrounding air by convection.
- The temperature distribution at different points on the slab cannot be directly observed and must be measured experimentally.
- This experiment focuses on measuring temperature at multiple locations on the slab to understand how heat spreads and how the surrounding air affects the cooling of the surface under convection conditions.
- At the end, we want to compare the simulation results with our experiment finding.

Working Principle

- Constant Temperature is maintained at one edge through a water bath.
- Heat spreads through the plate via conduction and convection at the edges.
- Energy balance at steady state governed by:
 $\nabla^2 T = 0$ (steady state)
- Boundary conditions:
 1. $T(x=0) = \text{constant}$
 2. Heat loss via convection

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Innovation in This Project

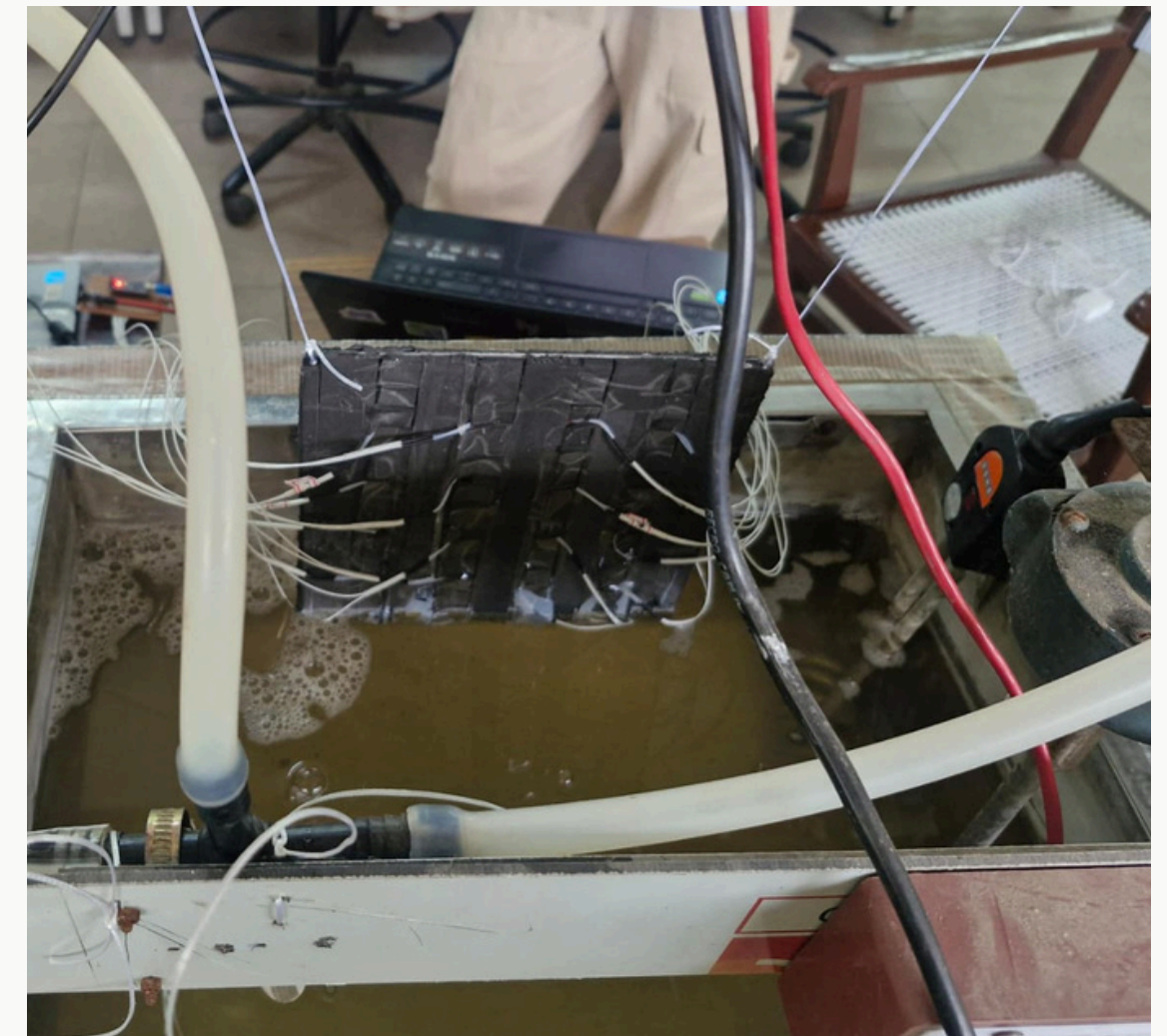
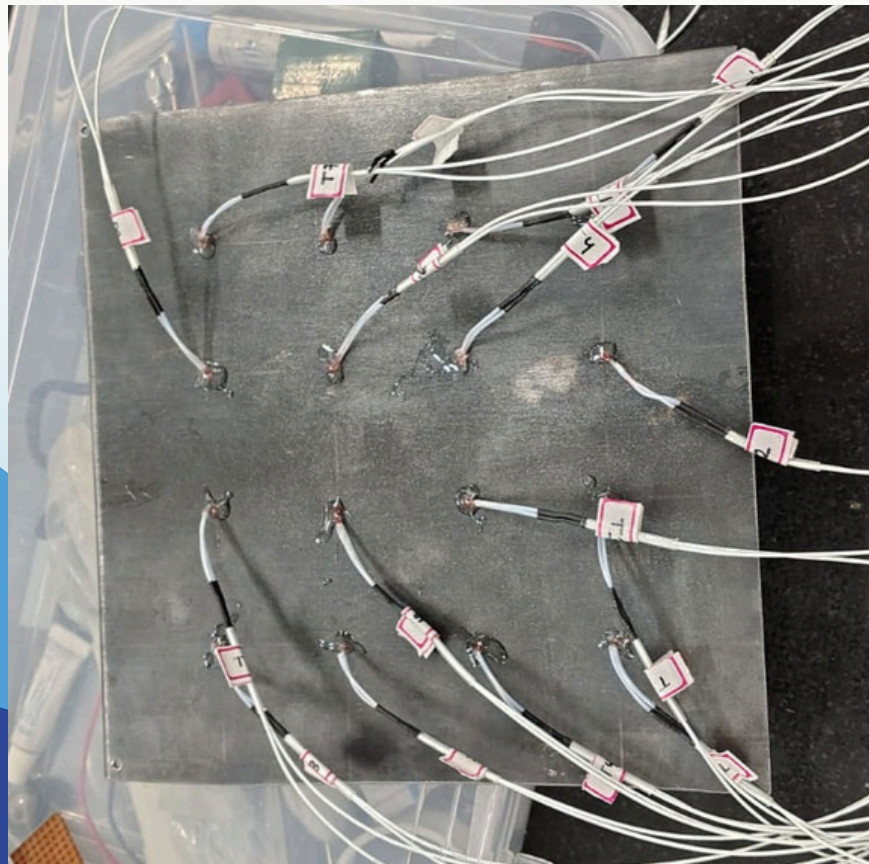
This project combines:

- Numerical simulation using MATLAB
- Multi-point temperature measurement
- Convection in ambient Surrounding.
- Direct comparison of theory and experiment.

This provides a complete heat transfer visualization system.

Experimental Setup

1. 20cm x 20cm square stainless steel
2. 16 Temperature sensors
3. Insulation Tape
4. Temperature controlled bath
5. Laptop for recording data

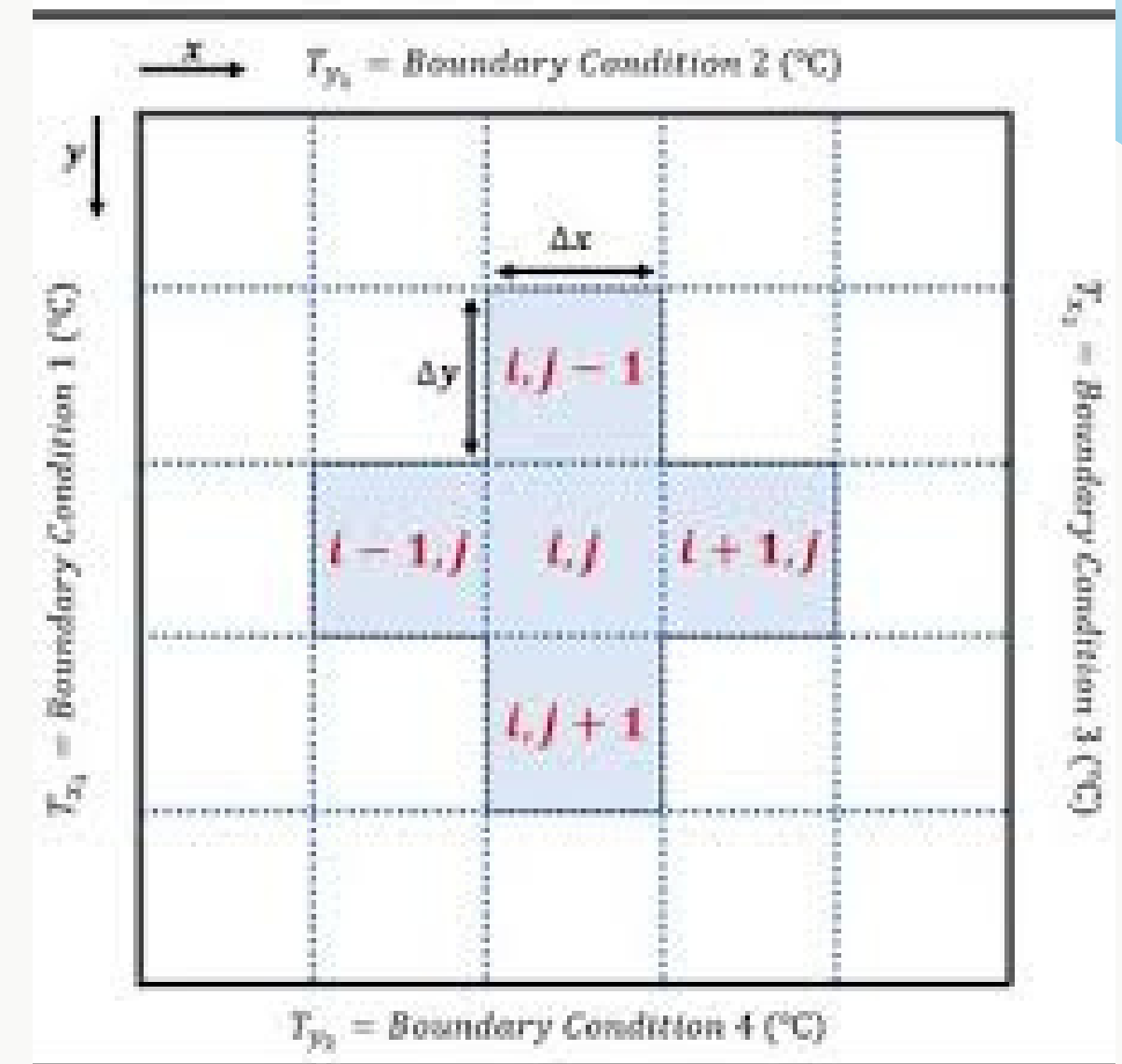


Experimental Video and Set Up



Experimental Procedure

1. The metal slab (20cm by 20cm) was insulated from the bottom and top surfaces.
2. Temperature sensors were attached at sixteen different points on the slab to measure temperature distribution.
3. All temperature sensors were connected to the data acquisition system and computer.
4. One edge of the slab was kept at a constant temperature
5. Once kept properly, heat conduction starts from one end and spreads to the other sides of the plate.



Experimental Procedure

6. The temperature profile starts to develop gradually across the entire plate.
7. Once the steady state is reached, we turn off the water bath.
8. Temperature readings were then continuously recorded as the slab cooled naturally due to natural convection to the surrounding air.
9. The system was observed until the temperature approached steady ambient conditions.
10. The collected temperature data during both heating and cooling were used to analyse the temperature distribution in the slab.

Physical Assumptions

1. There is no variations in temperature in Z dir of the plate since the plate is very thin.



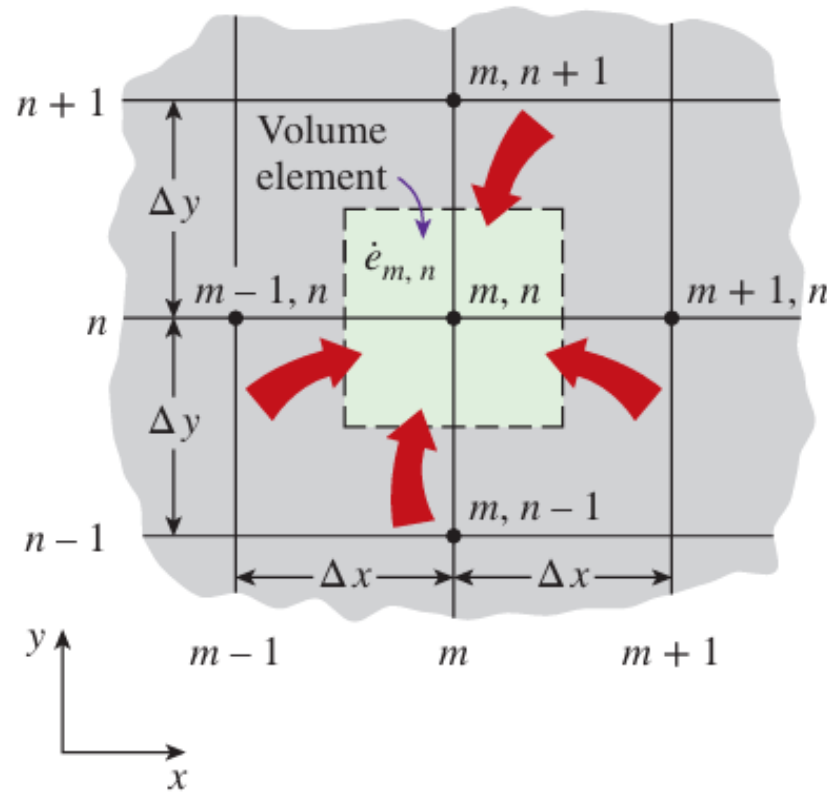
2. Proper Insulation from bottom and top of the surface. Hence no convection heat loss from the two surfaces.



Finite Difference Method

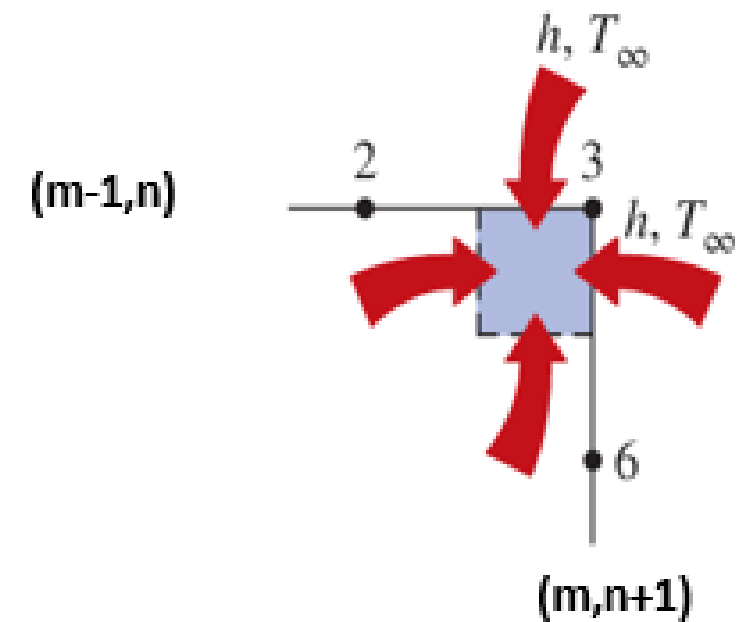
Interior Node

$$T_{m-1,n} + T_{m+1,n} + T_{m,n+1} + T_{m,n-1} - 4T_{m,n} = 0$$



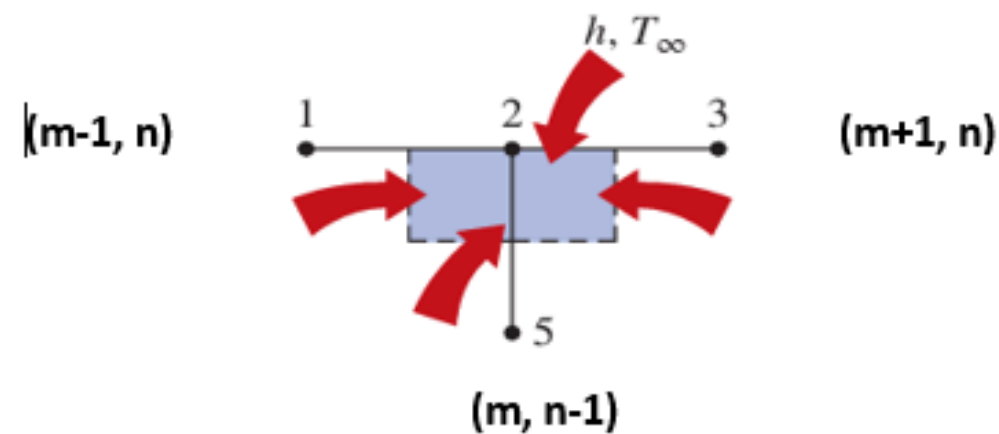
Corner Node

$$h \left(\frac{\Delta x}{2} + \frac{\Delta y}{2} \right) (T_{\infty} - T_{m,n}) + k \frac{\Delta x}{2} \frac{T_{m,n-1} - T_{m,n}}{\Delta y} + k \frac{\Delta y}{2} \frac{T_{m-1,n} - T_{m,n}}{\Delta x} = 0$$



Left Boundary Node

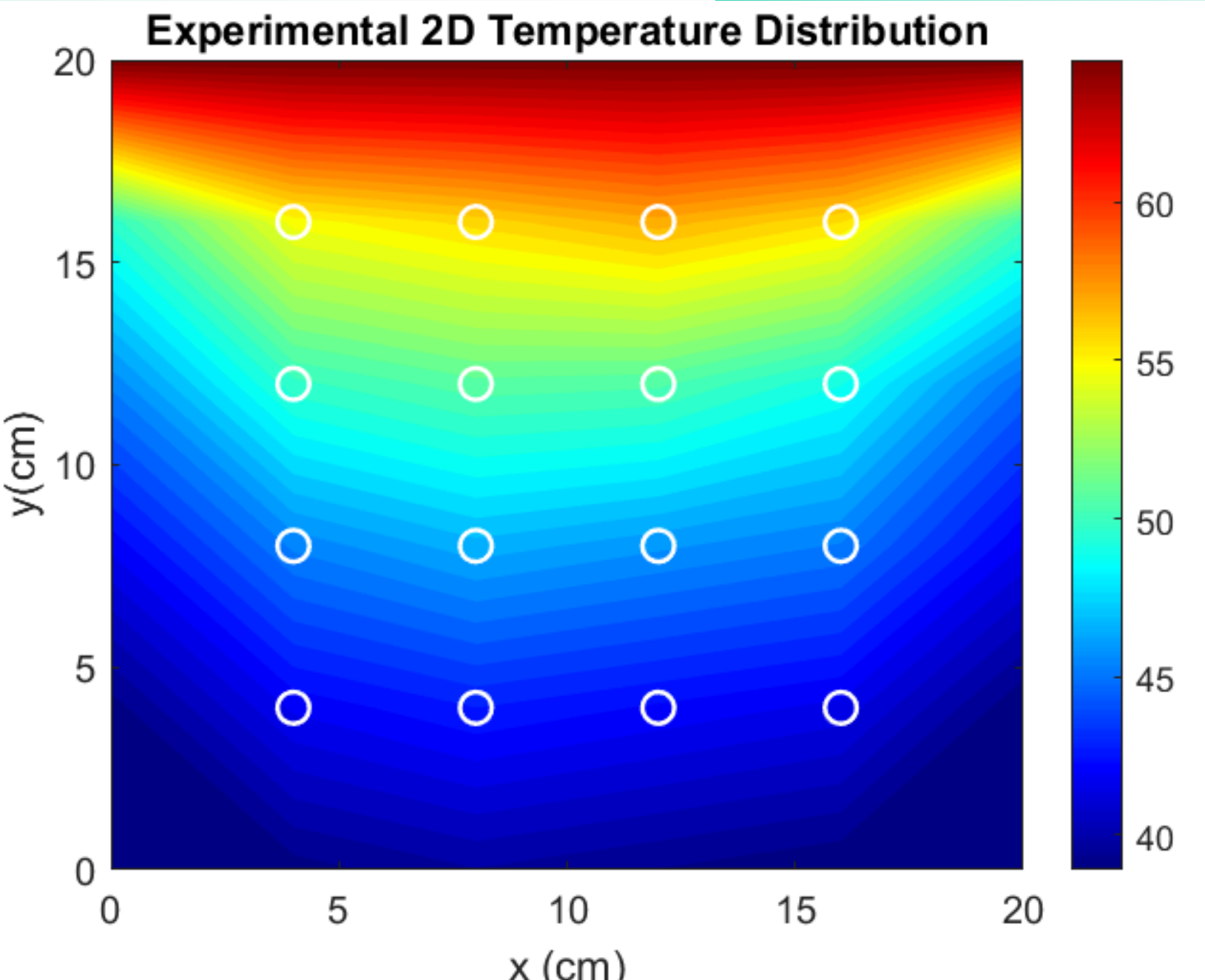
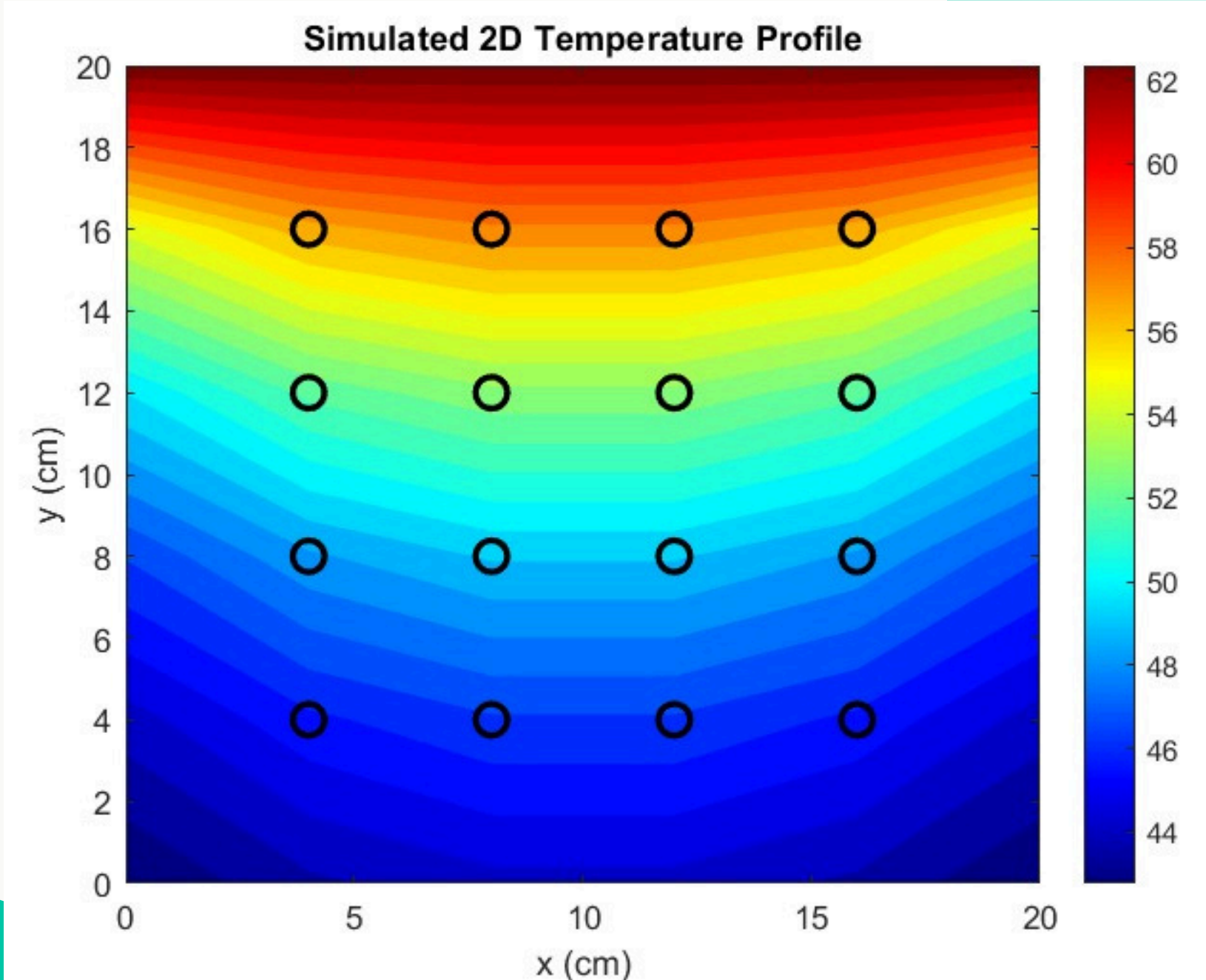
$$h\Delta x(T_{\infty} - T_{m,n}) + \frac{k\Delta y}{2} \frac{T_{m+1,n} - T_{m,n}}{\Delta x} + k\Delta x \frac{T_{m,n-1} - T_{m,n}}{\Delta y} + \frac{k\Delta y}{2} \frac{T_{m-1,n} - T_{m,n}}{\Delta x} = 0$$



Experimental vs Simulated Temperature(@ Sensors Locations)

Location (x,y) [cm]	Experimental [°C]	Simulated [°C]	Rel. Error (%)
(4, 4)	41.8	45.8807	9.76%
(8, 4)	42.4	46.602	9.91%
(12, 4)	43	46.602	8.38%
(16, 4)	42.1	45.8808	8.98%
(4, 8)	45.5	48.532	6.66%
(8, 8)	46.2	49.3927	6.91%
(12, 8)	46.9	49.3927	5.31%
(16, 8)	45.7	48.532	6.20%
(4, 12)	49.2	52.1029	5.90%
(8, 12)	52.1	53.0442	1.81%
(12, 12)	51.8	53.0443	2.40%
(16, 12)	50.2	52.1029	3.79%
(4, 16)	55	56.867	3.39%
(8, 16)	56	57.6371	2.92%
(12, 16)	57.1	57.6371	2.73%
(16, 16)	55.8	56.867	1.91%

2-D Surface Plot (Simulated vs Experimental)



Code Implementation

Interior Nodes

```
% Update Interior Nodes |
for i = 2:N-1
    for j = 2:N-1
        T(i,j) = 0.25 * (T(i+1,j) + T(i-1,j) + T(i,j+1) + T(i,j-1));
    end
end
```

Corner Nodes

```
% Corner Nodes (Special Case of Convection Edges)
T(1, 1) = (T(2, 1) + T(1, 2) + 2*Bi*Tinf) / (2 + 2*Bi);
T(1, N) = (T(2, N) + T(1, N-1) + 2*Bi*Tinf) / (2 + 2*Bi);
```

Dirichlet Boundary Condtion

```
% BC 4: TOP (i=N) Edge (Dirichlet - Water Bath)
% This is the 4th BC. We override the top row with 65C.
T(N, :) = Th;
```

Boundary Left and Right Edge Nodes

```
% BC 1 & 2: LEFT (j=1) and RIGHT (j=N) Edges (Convection/Robin)
% Using the rigorous Control Volume balance from your derivation
for i = 2:N-1
    T(i, 1) = (0.5*(T(i+1, 1) + T(i-1, 1)) + T(i, 2) + Bi * Tinf) / (2 + Bi);
    T(i, N) = (0.5*(T(i+1, N) + T(i-1, N)) + T(i, N-1) + Bi * Tinf) / (2 + Bi);
end
```

Bottom Edge(x=L)

```
% BC 3: BOTTOM (i=1) Edge (Convection/Robin)
for j = 2:N-1
    T(1, j) = (0.5*(T(1, j+1) + T(1, j-1)) + T(2, j) + Bi * Tinf) / (2 + Bi);
end
```



**Now we will show our
simulated model in
MATLAB**

Sources of Error

1. Limited number of sensors: We had only 16 sensors for a plate area of 400 cm^2 ; increasing the number will decrease error.
2. Fitted "h" value: We used the convective heat transfer coefficient(h), which fits our experimental data
3. No perfect insulation: Even though it has insulation on the top and bottom surface its not 100% insulated.
4. Water bath temperature varied 1-2 degrees Celsius.
5. "Non-Ideal Boundary Contact: Incidental immersion of the lateral faces created a volumetric heat source rather than a purely linear edge contact
6. Heat loss due to radiation is neglected

Conclusion

1. **We** see that the experimental and simulated results are similar. However errors are still there in the two results.
2. Numerical methods works well in cases where finding experimental results are difficult. Finite difference method is good choice to simulate the system. Its accuracy can be increase by increasing the number of nodes.
3. Fluid flow over the edges of the plate makes the use of appropriate “h” complex. Its constitute the air as well the water vapor fumes rising from the water bath.
4. Best fitting the experimental data with the simulation gives us the approriate results to be $h \sim 78 \text{ W}/(\text{m}^2.\text{K})$.

Revised Content

Modification in Assumptions

- 1. Our lateral surfaces is no longer perfectly insulated.**
- 2. It is subjective to heat losses through convection and radiation.**
- 3. Heat from lateral surface goes through the insulation material through conduction and ultimately to ambient by convection.**

Changes

1.Face Loss Formula (Insulation Leakage)

In the previous model, we assumed q loss from the face = 0. In the current model, we account for the heat conduction through the insulation on both sides of the plate.

2. Radiation Loss Formula

We added the Stefan-Boltzmann law to account for the energy emitted by the plate's surface as infrared radiation.

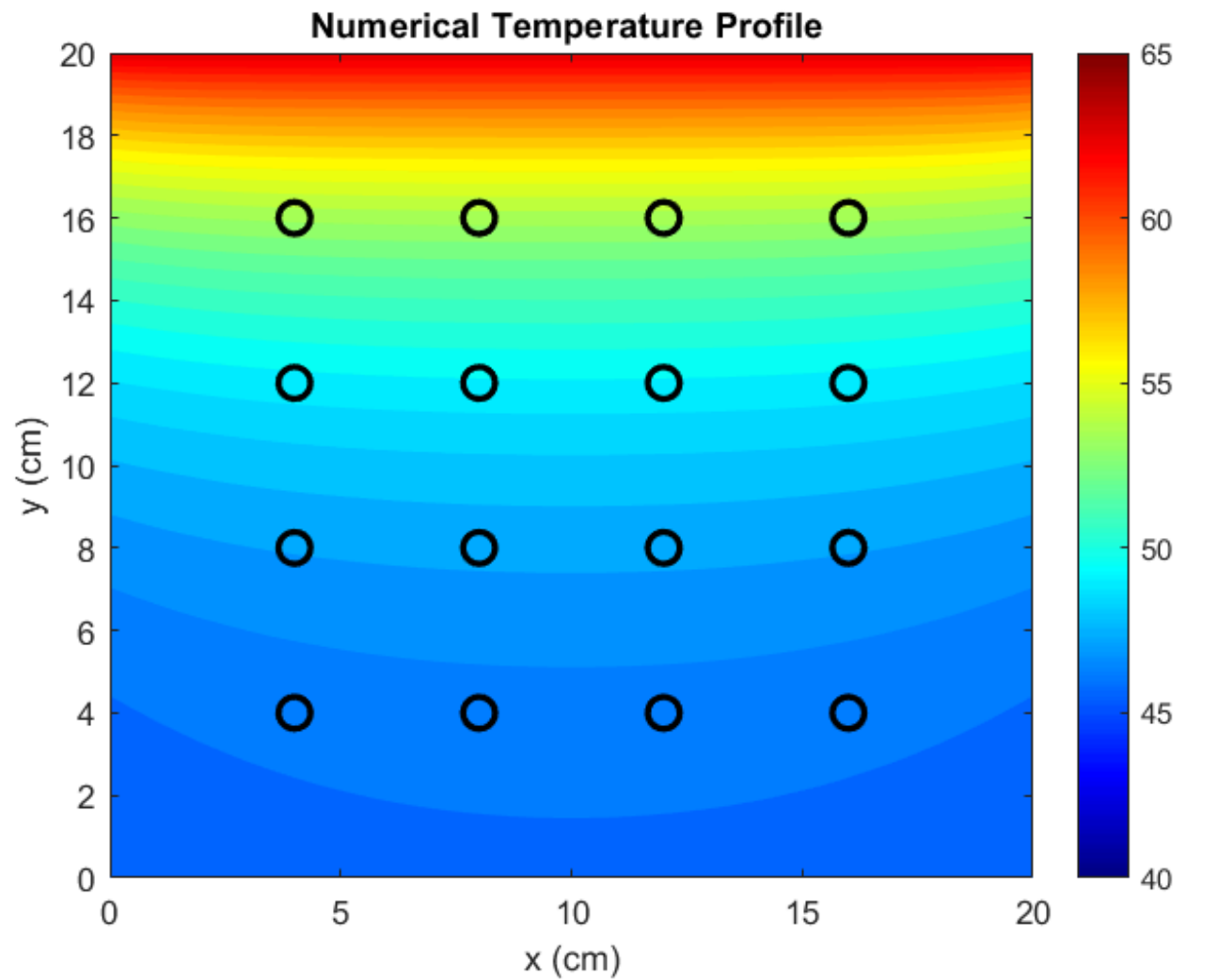
$$q_{rad} = \epsilon \cdot \sigma \cdot (T_K^4 - T_{sur,K}^4)$$

$$h_{rad} = \epsilon \cdot \sigma \cdot (T_K + T_{sur,K}) \cdot (T_K^2 + T_{sur,K}^2)$$

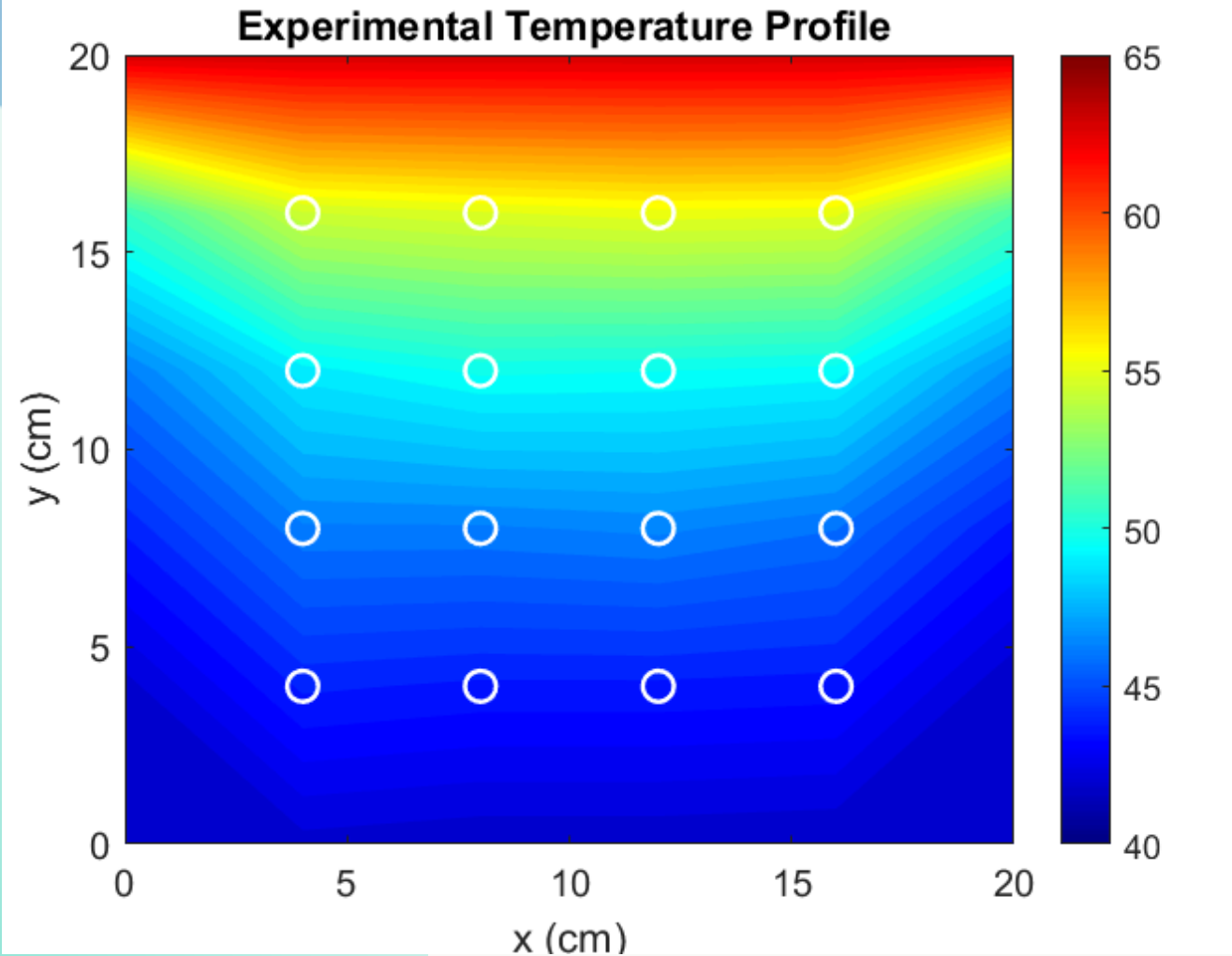
3. When you combine conduction, face loss, and radiation into a single energy balance for an interior node, the Finite Difference Equation changes from a simple average to this weighted version:

$$T_{i,j} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} + (loss_f \cdot T_{s,avg})}{4 + loss_f}$$

Contour Plot (Simulated vs Experimental)

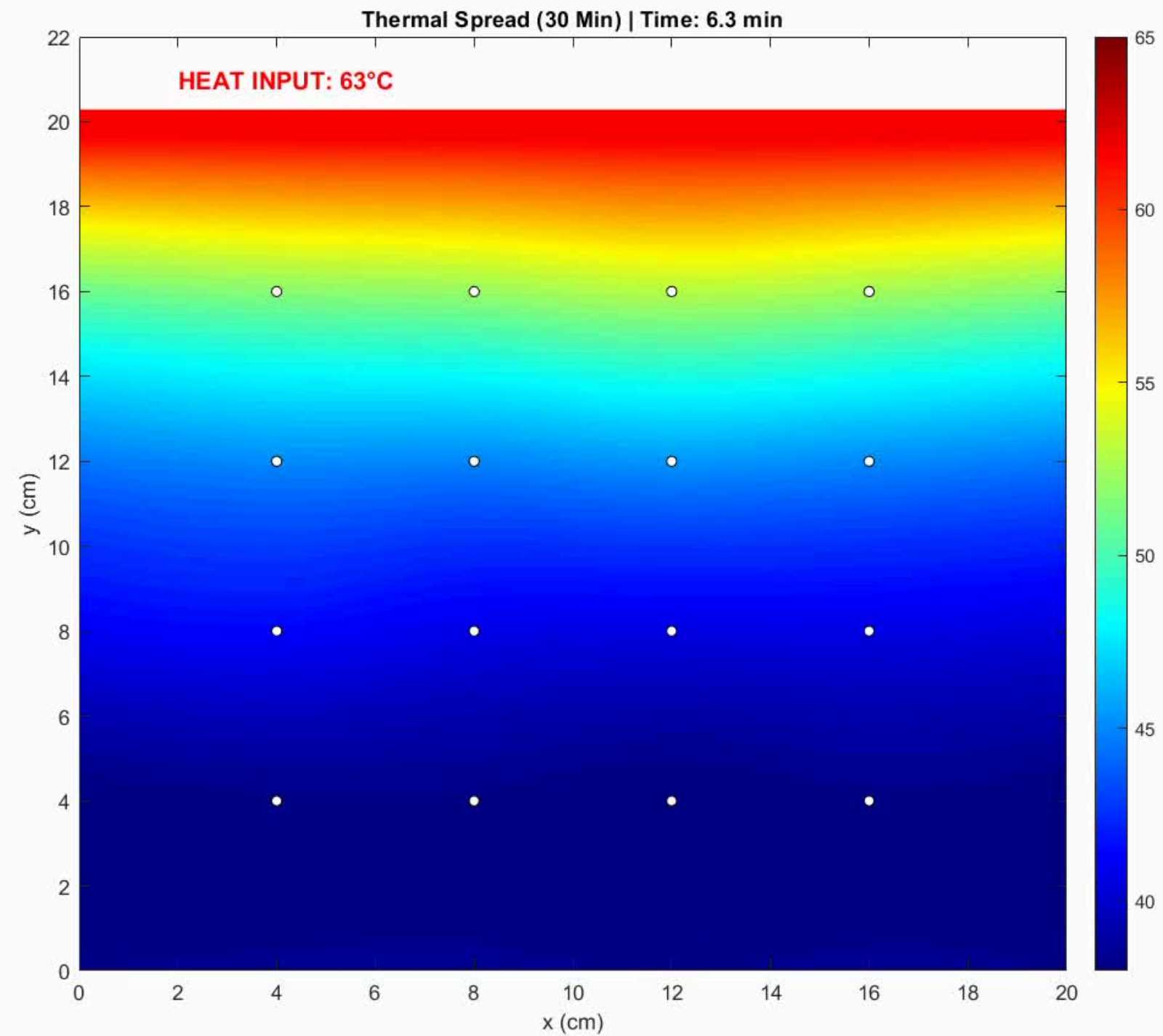


Simulated



Experimental

Animation of Temperature Profile Developing across the plate



Experimental vs Simulated Temperature(@ Sensors Locations)

Location (x, y) cm	Experimental (°C)	Simulated (°C)	Percentage Error (%)
(4,4)	44.1	46.4014	5.22
(8,4)	43.9	46.5151	5.96
(12,4)	43.9	46.5151	5.96
(16,4)	43.8	46.4014	5.94
(4,8)	46.4	47.3953	2.15
(8,8)	46.4	47.5203	2.41
(12,8)	46.6	47.5203	1.97
(16,8)	46.1	47.3953	2.81
(4,12)	49.2	49.1638	0.07
(8,12)	49.9	49.299	1.2
(12,12)	49.8	49.299	1.01
(16,12)	49.6	49.1638	0.88
(4,16)	54.7	53.4585	2.27
(8,16)	55.1	53.5777	2.76
(12,16)	55.4	53.5777	3.29
(16,16)	55.3	53.4584	3.33

Conclusions

1. Our assumption of proper insulation is highly deviating from the reality.
2. Proper Insulation: $h \sim 70 \text{ W}/(\text{m}^2.\text{k})$
Improper Insulation: $h \sim 10 \text{ W}/(\text{m}^2.\text{k})$
3. Experimental data shows us the evolution of temperature profile over the plate with time.
4. Acknowledging the heat losses from the back and front due to lack of proper insulation then we are able to get a reasonable value of $h \sim 10 \text{ W}/(\text{m}^2.\text{k})$.



**THANK
YOU!!**